

Between-Subject Variability in Memory Modulation

Patient-level linear models using the behavioral first-session summary file

Requested outcome: avg_dprime_diff. The behavioral file does not contain a column with that exact name; the available patient-level outcome is avg_stim_dprime_diff, which was used here.

Critical modeling point: this file has one row per patient (54 rows, 54 unique patients). Because the outcome is already collapsed to a single number per person, a subject random intercept is not identifiable. The correct analysis at this level is a patient-level linear model, not an MLM with (1 | subject).

Executive Summary

- Unadjusted sex difference: males $M = 0.179$, females $M = -0.039$, mean difference = 0.218, Welch $t(42.19) = 2.781$, $p = .008$.
- Primary adjusted model: $R^2 = 0.205$, adjusted $R^2 = 0.140$, $F(4, 49) = 3.154$, $p = .022$.
- Among the requested between-subject predictors, sex showed the largest unique contribution in the primary model, but it remained marginal rather than conventionally significant after adjustment.
- IED_freq and Memory_Z did not improve model fit in the smaller subsets where they were available.

Interpretive frame

These models do not explain the original trial-level ICC directly. Instead, they test whether the requested patient-level covariates are associated with between-subject differences in the already-collapsed memory modulation score.

Data Availability and Descriptives

Table 1
Variable availability in the behavioral summary file

Variable	Label	Non-missing	Missing	Percent available
avg_stim_dprim	Memory modulation (avg_stim_dprime_diff)	54	0	100.0%
sex	Sex (male vs female)	54	0	100.0%
age	Age (years)	54	0	100.0%
stim_hemispher	Stim hemisphere (R vs L)	54	0	100.0%
stim_DB	Stim amplitude (DB)	54	0	100.0%
IED_freq	IED frequency (Likert)	44	10	81.5%
Memory_Z	Pre-surgical baseline memory (Memory_Z)	31	23	57.4%

Note. Memory_Z is available for 31/54 patients and is concentrated in the Original, Timing, and Duration cohorts. IED_freq is available for 44/54 patients.

In the raw patient-level outcome, males showed higher avg_stim_dprime_diff than females. That unadjusted difference is useful descriptively, but the primary question is whether it survives after accounting for age, stimulation hemisphere, and stimulation amplitude.

Table 2
Outcome descriptives by sex

Group	N	Mean	SD	Median	Min	Max
female	34	-0.039	0.290	-0.053	-0.526	0.840
male	20	0.179	0.271	0.230	-0.779	0.440

Note. Means are computed on avg_stim_dprime_diff. Positive values indicate greater memory performance on stimulated relative to non-stimulated trials.

Memory_Z subset composition: Original (n=14), Timing (n=12), Duration (n=5). Within that subset, stim_DB has no variation: all available cases are 0.5 DB.

Primary Full-Sample Model

Primary patient-level model using the full cohort. This is the cleanest adjusted test because all four predictors are available for all 54 patients.

Table 3
Primary model fit summary

N	Predictors	R2	Adj R2	F(df)	Model p
54	4	0.205	0.140	3.154 (4, 49)	.022

Note. The primary model includes sex, age, stimulation hemisphere, and stimulation amplitude.

Table 4
Primary model coefficients

Term	B	SE	t	p	95% CI
Intercept	-0.111	0.172	-0.643	.523	[-0.457, 0.235]
Sex (male vs female)	0.160	0.085	1.887	.065	[-0.010, 0.331]
Age (years)	0.006	0.004	1.616	.113	[-0.001, 0.013]
Stim hemisphere (R vs L)	-0.122	0.081	-1.509	.138	[-0.285, 0.041]
Stim amplitude (DB)	-0.091	0.181	-0.505	.616	[-0.454, 0.272]

Note. Female and left hemisphere are the reference groups for binary predictors.

The primary adjusted model was significant overall, $F(4, 49) = 3.154$, $p = .022$, and accounted for 0.205 of the variance in avg_stim_dprime_diff. The adjusted coefficient for sex was $B = 0.160$ (95% CI [-0.010, 0.331]), $t = 1.887$, $p = .065$. That is smaller than the raw male-female difference of 0.218, which indicates part of the unadjusted sex effect overlaps with the other covariates.

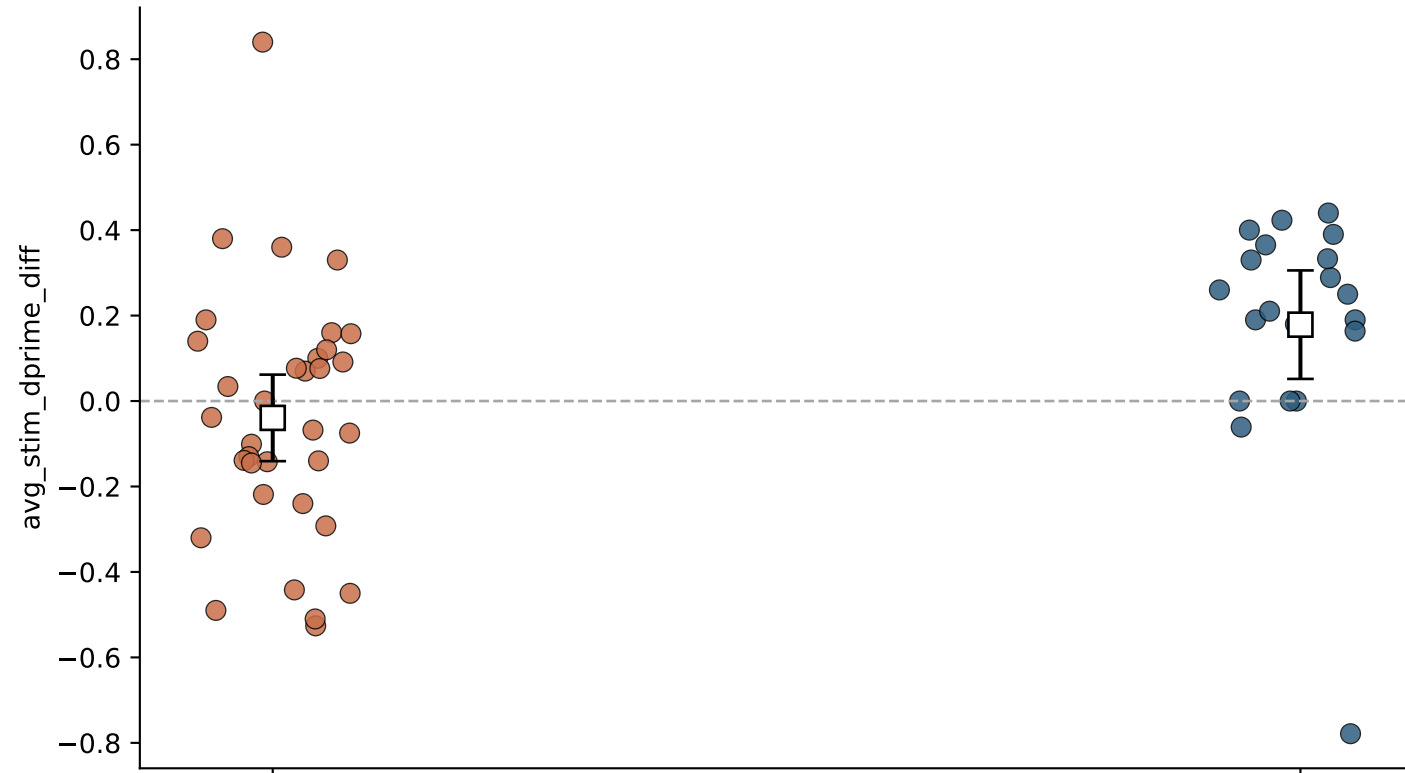
Table 5
Unique variance contribution in the primary model

Predictor	Delta R2	F change	p
Sex (male vs female)	0.058	3.560	.065
Age (years)	0.042	2.610	.113
Stim hemisphere (R vs L)	0.037	2.277	.138
Stim amplitude (DB)	0.004	0.255	.616

Note. Delta R2 values reflect the drop in model R2 when each predictor is removed from the primary model.

Unadjusted Sex Difference

Raw patient-level outcome by sex



Welch's test on the raw patient-level outcome gave $t(42.19) = 2.781$, $p = .008$, with males higher than females by 0.218 (95% CI [0.060, 0.377]). The adjusted primary model retained sex as the largest unique contributor, but not at $p < .05$.

Sensitivity Models

The sensitivity models show what happens when IED_freq and Memory_Z are brought into the analysis. The cost is lower N and, for Memory_Z, a narrower set of study cohorts. Within those reduced subsets, no additional requested predictor produced a clear signal.

Table 6
Comparison of fitted patient-level models

Model	N	Predictors used	Omitted in subset	R2	Adj R2	Model p
Primary Full-Sampl	54	Sex (male vs female); Age (years)		0.205	0.140	.022
IED Frequency Se	44	Sex (male vs female); Age (years)		0.163	0.053	.219
Memory_Z Sensiti	31	Sex (male vs female); Age (years)	Stim amplitude (DB) (no va	0.128	-0.007	.451
Complete-Case Exp	22	Sex (male vs female); Age (years); Stim amplitude (DB) (no va	IED frequency (Hz) (no va	0.146	-0.006	.387

Note. In the Memory_Z-only and complete-case subsets, stim_DB was automatically omitted because all available cases were at 0.5 DB, leaving no within-subset variation to estimate its effect.

IED Frequency Sensitivity Model

Sensitivity model adding IED frequency. This costs 10 patients because IED_freq is missing for part of the sample.

Table
IED Frequency Sensitivity Model coefficients

Term	B	SE	t	p	95% CI
Intercept	-0.187	0.213	-0.876	.386	[-0.619, 0.245]
Sex (male vs female)	0.146	0.106	1.380	.176	[-0.068, 0.360]
Age (years)	0.007	0.005	1.385	.174	[-0.003, 0.016]
Stim hemisphere (R vs L)	-0.106	0.097	-1.089	.283	[-0.303, 0.091]
Stim amplitude (DB)	-0.072	0.210	-0.341	.735	[-0.496, 0.353]
IED frequency (Likert)	0.023	0.062	0.371	.713	[-0.102, 0.148]

Note. Female and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 44, R2 = 0.163, adjusted R2 = 0.053, F(5, 38) = 1.481, p = .219. Omitted predictors: None.

Table
IED Frequency Sensitivity Model unique variance contribution

Predictor	Delta R2	F change	p
Sex (male vs female)	0.042	1.905	.176
Age (years)	0.042	1.918	.174
Stim hemisphere (R vs L)	0.026	1.185	.283
Stim amplitude (DB)	0.003	0.117	.735
IED frequency (Likert)	0.003	0.137	.713

Note. Delta R2 values are specific to the analysis subset for that model.

Memory_Z Sensitivity Model

Sensitivity model adding pre-surgical baseline memory. The available sample is restricted to the studies where Memory_Z exists.

Table
Memory_Z Sensitivity Model coefficients

Term	B	SE	t	p	95% CI
Intercept	-0.107	0.163	-0.656	.518	[-0.442, 0.228]
Sex (male vs female)	0.103	0.110	0.933	.359	[-0.123, 0.328]
Age (years)	0.005	0.005	1.118	.274	[-0.004, 0.014]
Stim hemisphere (R vs L)	-0.073	0.126	-0.578	.568	[-0.333, 0.187]
Pre-surgical baseline memory (Memory_Z)	-0.054	0.069	-0.781	.442	[-0.195, 0.087]

Note. Female and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 31, R2 = 0.128, adjusted R2 = -0.007, F(4, 26) = 0.951, p = .451. Omitted predictors: Stim amplitude (DB) (no variation in analysis subset).

Table
Memory_Z Sensitivity Model unique variance contribution

Predictor	Delta R2	F change	p
Sex (male vs female)	0.029	0.871	.359
Age (years)	0.042	1.249	.274
Stim hemisphere (R vs L)	0.011	0.334	.568
Pre-surgical baseline memory (Memory_Z)	0.020	0.610	.442

Note. Delta R2 values are specific to the analysis subset for that model.

Complete-Case Exploratory Model

Exploratory complete-case model including all requested between-subject predictors. This is the most data-limited model and is shown mainly as a sensitivity check.

Table
Complete-Case Exploratory Model coefficients

Term	B	SE	t	p	95% CI
Intercept	-0.156	0.237	-0.660	.519	[-0.658, 0.345]
Sex (male vs female)	0.060	0.168	0.359	.725	[-0.295, 0.416]
Age (years)	0.005	0.008	0.635	.534	[-0.011, 0.021]
Stim hemisphere (R vs L)	-0.001	0.187	-0.005	.996	[-0.397, 0.395]
IED frequency (Likert)	0.035	0.084	0.414	.684	[-0.144, 0.213]
Pre-surgical baseline memory (Memory_Z)	-0.075	0.106	-0.707	.490	[-0.301, 0.150]

Note. Female and left hemisphere are the reference groups when present. Constant predictors in a subset were omitted automatically.

N = 22, R2 = 0.111, adjusted R2 = -0.166, F(5, 16) = 0.402, p = .841. Omitted predictors: Stim amplitude (DB) (no variation in analysis subset).

Table
Complete-Case Exploratory Model unique variance contribution

Predictor	Delta R2	F change	p
Sex (male vs female)	0.007	0.129	.725
Age (years)	0.022	0.404	.534
Stim hemisphere (R vs L)	0.000	0.000	.996
IED frequency (Likert)	0.010	0.171	.684
Pre-surgical baseline memory (Memory_Z)	0.028	0.500	.490

Note. Delta R2 values are specific to the analysis subset for that model.

Bottom Line

This file supports a between-subject regression question, not a within-subject MLM question. The outcome is already one number per patient, so there is no residual subject-level nesting to model.

In the full-sample adjusted model, sex was the strongest requested predictor of memory modulation: $B = 0.160$, 95% CI $[-0.010, 0.331]$, $p = .065$. That effect was suggestive but not conventionally significant after adjustment.

IED_freq did not contribute meaningfully in the 44-patient sensitivity model (model $p = .219$), and adding it did not improve explanatory power.

Memory_Z was only available for 31 patients from a restricted subset of studies, and its coefficient was not significant in that subset (model $p = .451$). Because all cases in that subset were at 0.5 DB, stim_DB could not be estimated there.

If the substantive question is still what explains the high ICC from the original trial-level null model, the next step is to return to the trial-level dataset and add these patient-level covariates there as fixed effects in a mixed model. The present report instead shows which patient-level covariates relate to the collapsed patient summary outcome.

Recommended interpretation

Among the requested subject-level variables, sex looks like the most plausible contributor to between-subject differences in avg_stim_dprime_diff, but the adjusted evidence is modest. Age and hemisphere show smaller nonsignificant trends. Neither stimulation amplitude nor IED_freq explained much variance in these patient-level models.